Have Newer Models Learned to Read Idioms?

A Re-examination of Idiomaticity in Word Representations, from 2024 Encoders to 2026 LLMs, Calibrated Against Ordinary Perturbations

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Abstract

An idiom like *grey matter* means "brain," not "material that is grey." He et al. (2025) showed that 2024-era word-representation models do not encode such holistic meanings: when the compound is swapped for its gold synonym, models often find the literal pieces (*matter*, *silvery material*) closer than the correct paraphrase. We ask whether the rapid progress since then has changed this. We rebuild their minimal-pair probe as a model-agnostic framework and run it on twenty models: multilingual encoders, the strongest 2025 and 2026 sentence-embedding models, and a decoder-only LLM cohort (Qwen3, Qwen3.5, Gemma 3, Gemma 4; base and instruction-tuned; 0.8B to 12B), on English and Portuguese. To run linear-attention and multimodal LLMs on Apple Silicon we add an MLX path and a layer-recording recipe that we show is essential for these models. We find a soft ceiling. Every model scores below the partial-capture threshold; the best, base Gemma 4-12B, reaches only 0.525 while becoming the first model to actually prefer the gold synonym over the literal one. We then add the control the original protocol lacks: the same synonym-versus-random test on ordinary, non-idiomatic words (*doctor* to *physician* vs *carpet*) and phrases. No model separates idiomatic synonyms from random replacements as well as it does for ordinary targets. This calibration is the strongest evidence that the failure is specific to idiomaticity, not a generic insensitivity of embeddings, and that scale and newer generations approach but do not cross the bar.

1 Introduction

The problem in one sentence. A multiword expression is *idiomatic* when its meaning cannot be recovered from its parts. *Grey matter* is the brain; *red tape* is bureaucracy; a *cash cow* is a reliable source of money. A model that truly "understands" such an expression should place it, in its representation space, near its real meaning (near *brain*, *bureaucracy*, *reliable earner*) and far from a literal re-reading of its words. The question of this paper is simple to state: do today's models do this, and have the last two years of progress helped?

Why this is not obvious to test. You cannot just ask a model "is *grey matter* idiomatic?," because that tests its chat behaviour, not its representations. He et al. [1] instead use a *minimal-pair probe*: take a sentence containing the compound, replace the compound with carefully chosen substitutes, and measure how much the representation moves. If replacing *grey matter* with *brain* (its correct meaning) barely changes the representation, but replacing it with *matter* (a literal piece) changes it a lot, the model has understood the idiom. Their finding was negative: 2024 models behaved the opposite way. We revisit that finding with newer and larger models, and we add a control that the original protocol was missing.

Contributions. This paper makes four contributions.

1. A model-agnostic re-implementation of the NCIMP probe and its metrics, with an MLX backend that lets linear-attention (Qwen3.5) and multimodal (Gemma 4) LLMs run on Apple Silicon, and a layer-recording recipe we show is necessary to get a fair representation from decoder LLMs (Section 4).

2. Four experiments under one protocol: replicating the paper on 2024 encoders in two languages (Section 5); 2025 and 2026 sentence-embedding models (Section 6); an eleven-model decoder-LLM cohort (Section 7); and an ordinary-perturbation control on all twenty models (Section 8).

3. A single, calibrated answer: a soft ceiling around 0.50 to 0.53 that nothing crosses, with the failure shown to be specific to idiomaticity rather than a generic embedding artefact (Section 10).

4. Two methodological findings that matter for anyone probing modern LLMs: the final-layer residual stream of decoder LLMs must be normalised before it can be measured, and the composite capture score must be read together with an anisotropy diagnostic.

Section 2 introduces the probe on a single sentence and Section 3 the metrics; Section 4 describes the framework; the four experiments follow; and Section 9 collects all twenty models in one table before the discussion.

2 The Probe, by Example

We introduce the method on one concrete sentence before giving any formulas. Take the idiomatic compound *grey matter* in

> *"These youngsters use their grey matter when the presentation is right."*

The probe replaces the target compound (the *noun compound*, NC) with four kinds of substitute, each chosen to test a specific expectation.

- P_{Syn}, the **gold holistic synonym**: *grey matter* to *brain*. This preserves the real meaning, so similarity *should be high* regardless of how idiomatic the compound is.
- P_{Comp}, the **single most informative component**: *grey matter* to *matter*. A part of the compound. For an idiom this loses the meaning, so similarity *should be low*; for a compositional compound it may stay high.
- P_{WordsSyn}, a **word-by-word literal synonym**: *grey matter* to *silvery material*. Each word replaced by an out-of-context synonym. For an idiom this gives nonsense, so similarity *should be low*.
- P_{Rand}, a **frequency-matched random expression**: *grey matter* to *battlefront serviceman*. Unrelated; a lower-bound control, similarity *should be lowest*.

So the pattern an idiomaticity-aware model *should* produce is: P_{Syn} high, then (P_{Comp}, P_{WordsSyn}) middling, then P_{Rand} low. Table 1 shows what one real model (BGE-M3) actually does, next to a compositional compound for contrast.

Compound	Substitution (probe)	cos	What should happen
grey matter (idiomatic)	*brain* (P_{Syn})	0.55	highest, but here it is the lowest non-random
	matter (P_{Comp})	**0.78**	low, but here it is the highest
	silvery material (P_{WordsSyn})	0.64	low, but here it beats the gold synonym
	battlefront serviceman (P_{Rand})	0.47	lowest, ok
economic aid (compositional)	*financial assistance* (P_{Syn})	**0.81**	highest, ok
	aid (P_{Comp})	0.73	(allowed to be high)
	budgetary assistance (P_{WordsSyn})	0.76	(allowed to be high)
	(random) (P_{Rand})	0.47	lowest, ok

Table 1: The whole study in one table (BGE-M3, contextual-span level). For the compositional *economic aid*, the gold synonym is correctly the closest substitute (0.81). For the idiomatic *grey matter*, the model puts the literal component *matter* (0.78) and the word-by-word *silvery material* (0.64) closer than the correct *brain* (0.55): it reads the idiom as the sum of its parts.

The dataset (NCIMP). The compounds come from the Noun Compound Idiomaticity Minimal Pairs dataset [1]: 280 English and 180 Portuguese compounds, each labelled by humans on a 0 (fully idiomatic) to 5 (fully compositional) scale and placed in three natural sentences plus an uninformative "neutral" template. Table 2 lists representative items in each class. We use the human compositionality score, denoted Comp, as the gold standard, and we report results at the NC level (the contextual span of the compound), which the paper finds most diagnostic.

Class	Example compound	Example sentence
Idiomatic	*grey matter* (brain)	"... use their **grey matter** when the presentation is right."
Idiomatic	*eager beaver* (hard worker)	"Eric was being an **eager beaver** and left work late."
Partly comp.	*Dutch courage* (courage from drink)	"... go down to the pub to get some **Dutch courage**."
Compositional	*economic aid* (financial help)	"... giving Cuba **economic aid** and military support."
Compositional	*research lab* (research facility)	"Being part of a **research lab** provides fieldwork."

Table 2: Representative NCIMP items. Each compound is probed with the four substitutes of Section 2.

3 Metrics, Each With Its Example

All metrics build on cosine similarity between two representation vectors. We keep using the *grey matter* numbers from Table 1 ($P_{\text{Syn}} = 0.55$, $P_{\text{Comp}} = 0.78$, $P_{\text{WordsSyn}} = 0.64$, $P_{\text{Rand}} = 0.47$) so every definition is concrete.

Similarity (S). The raw cosine between the original NC representation and the probe-modified one, $S(P_i) = \cos(e(P_i), e(\text{NC}))$. *Example:* $S(P_{\text{Syn}}) = \cos(\text{grey matter}, \text{brain}) = 0.55$.

Affinity (A). Which of two probes is closer? $A(P_i, P_j) = S(P_i) - S(P_j)$. *Example:* $A(P_{\text{Syn}}, P_{\text{WordsSyn}}) = 0.55 - 0.64 = -0.09$. The negative sign says the model prefers the literal "silvery material" to the correct "brain," exactly the wrong preference.

Scaled similarity (S^R). Embedding spaces are anisotropic, so even random vectors can look similar and a raw 0.55 is hard to read. We rescale against the random floor,

$$S^R(P_i) = \frac{S(P_i) - S(P_{\text{Rand}})}{1 - S(P_{\text{Rand}})}.$$

Example: $S^R(P_{\text{Syn}}) = (0.55 - 0.47)/(1 - 0.47) = 0.15$. So the gold synonym sits only 15% of the way from "random" to "identical," meaning the holistic meaning is almost not recovered.

Per-model diagnostics. Averaging over the idiomatic compounds gives four readable numbers, each with a clear ideal direction (Table 3).

Metric	Meaning	Ideal
ISC	idiomatic synonym capture, $\overline{S^R(P_{\text{Syn}})}$ on idioms	high (near 1)
LOD	lexical-overlap dominance, $\overline{S^R(P_{\text{WordsSyn}}) - S^R(P_{\text{Syn}})}$	below 0
FLOOR	random-similarity floor, $\overline{S(P_{\text{Rand}})}$ (anisotropy)	diagnostic
ISC→ICS	composite capture score in $[0, 1]$	high; ≥ 0.55 partial, ≥ 0.70 captures

Table 3: The four diagnostics, illustrated on *grey matter*: ISC asks "is *brain* recovered?"; LOD is positive here ($P_{\text{WordsSyn}}\,0.64 > P_{\text{Syn}}\,0.55$), the component-bias failure; FLOOR asks "does *battlefront serviceman* also look similar?"; ICS rolls these into one score.

A caveat we will need (anisotropy breaks the composite). Because S^R divides by $1-$FLOOR, the composite becomes unstable as FLOOR approaches 1. *Example:* a model whose space has collapsed to FLOOR $= 0.95$ has a denominator of 0.05, so tiny noise is amplified and the indicators drift to "neutral," which the composite can mistake for a decent score. We meet exactly this case (instruction-tuned Gemma 4-12B in Section 7) and therefore always report FLOOR next to ICS.

4 The Model-Agnostic Framework

The probe and the metrics are held fixed; only the model changes. Each model is wrapped as an *embedder* exposing two operations: embed a whole sentence, and embed the target span within its sentence (mean-pooling the relevant sub-tokens over the last four layers, following the original protocol). Table 4 lists the embedder types; static vectors, transformer encoders, sentence-embedding models, and decoder-only LLMs all fit one interface, so the probe code never changes between a 2024 encoder and a 2026 LLM.

Embedder type	Runtime	Used for
transformer	HuggingFace/torch	BERT, mBERT, DistilBERT, XLM-R
sentence	sentence-transformers	mSBERT, BGE-M3, E5, Qwen3-Embedding
causal	HuggingFace/torch	Qwen3, Gemma 4-E2B (text path)
mlx_causal	Apple MLX (mlx-lm / mlx-vlm)	Qwen3.5, Gemma 3, Gemma 4-12B

Table 4: Pluggable embedder types. Adding a model is one registry line; the probe and metrics are untouched, which is what makes the cross-generation comparison fair.

Why an MLX path was needed. Two of the newest LLM families do not run well through the standard PyTorch path on Apple Silicon. Qwen3.5 uses linear attention whose fast kernels are CUDA-only, so PyTorch falls back to an unusably slow implementation; and Gemma 4 is a unified multimodal architecture (`gemma4_unified`) that the text-only loader rejects. We therefore added an MLX embedder that loads text LLMs via `mlx-lm` and multimodal ones via `mlx-vlm`, converting the base checkpoints on the fly so we keep full-precision base weights. On an M2 Max this turns a multi-hour or impossible run into minutes.

Why the last layer alone is not enough, and the fix. MLX exposes only the final hidden state cheaply, but for decoder LLMs the final-layer residual stream is pathologically anisotropic. *Example:* for base Gemma 3-12B, the cosine between two unrelated sentences, *"grey matter functioning here"* and *"supermarket city raining now,"* is 0.98 from the raw last-four layers; everything looks alike. The cause is a large shared component that the model's own final RMSNorm is designed to remove. Recording the last four decoder layers and then applying that final norm drops the same cosine to 0.56 and restores a usable representation. We use this last-four-layers-plus-final-norm recipe for all MLX models; it is the difference between a meaningful score and noise, and it is one of the two methodological findings of this paper.

5 Experiment 1: Replicating the Paper on 2024 Encoders

Setup. Six multilingual models in two cohorts: paper-era (mBERT, multilingual DistilBERT, mSBERT) and 2024-modern (XLM-R-large, BGE-M3, multilingual E5-large), on English (1124 instances) and Portuguese (720), at NC level.

Expected vs. actual. As in Section 2, the gold synonym should be the closest substitute for every class, while the literal probes should fall off for idioms. Table 1 already showed the failure on *grey matter* and the success on *economic aid* for one of these models. Aggregated by cohort (Table 5) and by model (Table 6), the picture is uniform: English improves modestly from the older to the newer cohort, Portuguese does not, and no model crosses 0.55.

Lang.	Cohort	ISC	LOD	ICS
EN	paper-era (mBERT, DistilBERT-ML, mSBERT)	0.099	0.227	0.341
EN	2024-modern (XLM-R, BGE-M3, E5-large)	0.174	0.077	0.430
PT	paper-era	0.150	0.203	0.428
PT	2024-modern	0.095	0.158	0.423

Table 5: Encoder cohorts at NC level (mean over naturalistic and neutral contexts). English gains modestly (0.341 to 0.430); Portuguese is flat. LOD stays positive throughout: the literal pieces beat the holistic synonym for idioms in every cohort.

	English				Portuguese			
Model	ISC	LOD	FLOOR	ICS	ISC	LOD	FLOOR	ICS
mBERT	0.085	0.205	0.777	0.353	0.125	0.191	0.795	0.435
DistilBERT-ML	0.021	0.181	0.744	0.349	0.004	0.191	0.779	0.419
mSBERT	0.191	0.295	0.205	0.320	0.322	0.228	0.250	0.431
XLM-R-large	0.100	0.071	0.935	0.424	-0.058	0.219	0.947	0.404
BGE-M3	0.229	0.069	0.430	0.428	0.216	0.055	0.417	0.449
E5-large	0.193	0.091	0.803	0.438	0.127	0.201	0.826	0.415

Table 6: Per-model encoder results. Note XLM-R-large's very high FLOOR (0.93 to 0.95): an extremely anisotropic space, where its low LOD is partly an artefact of everything being similar.

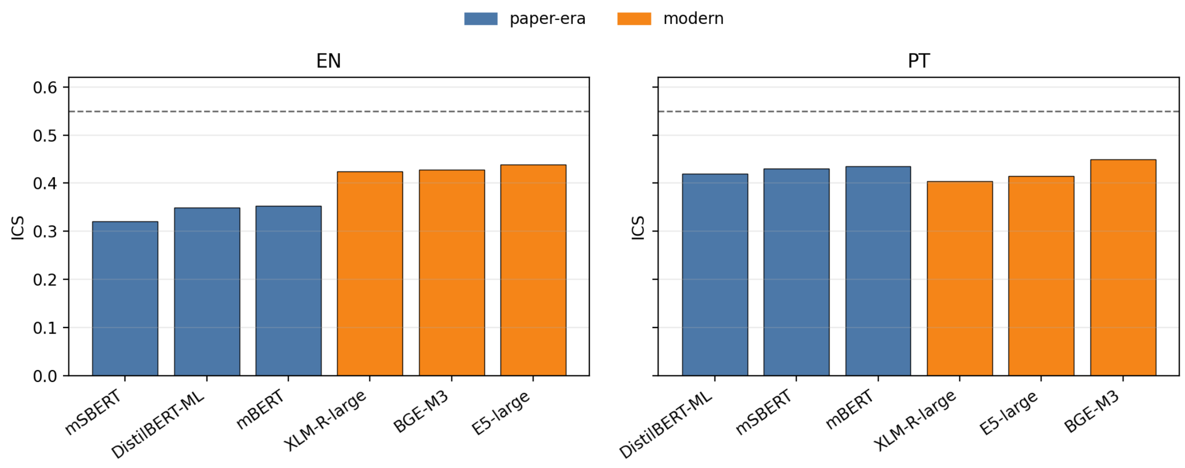

Figure 1: Per-model ICS by language. The dashed line is the partial-capture threshold (0.55); no model reaches it. The best English models (E5-large, BGE-M3) only modestly exceed the paper-era ones, and in Portuguese the ordering is essentially flat.

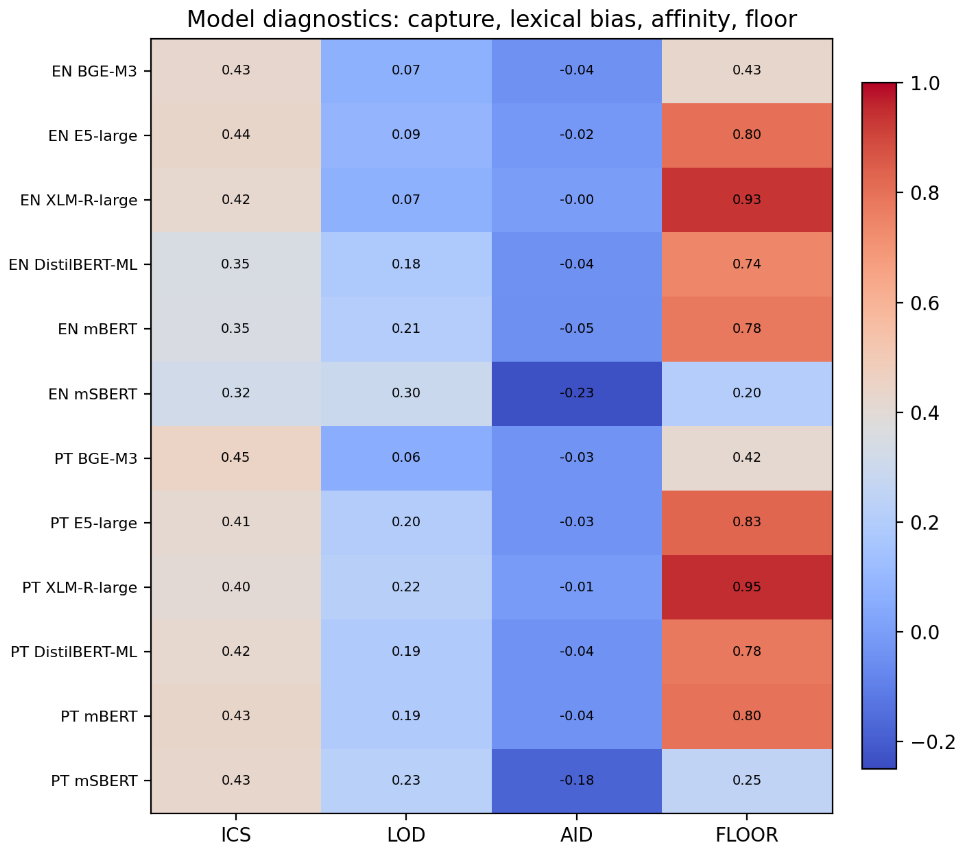

Figure 2: Diagnostic heatmap (capture, lexical bias, affinity, floor) for the encoder cohort. The high-FLOOR rows (XLM-R-large, E5-large) show how anisotropy and lexical bias interact: a low LOD under a near-1 floor is not the same as genuine idiom understanding.

Reading. This reproduces He et al.: 2024 models do not represent idiomaticity, and the improvement is language-dependent and small. That the English gain does not transfer to Portuguese hints it may come from English-heavy training data rather than from genuinely learning idioms. But these tables cannot yet separate "the model fails on idioms" from "the measurement is insensitive in general," which is what Experiment 4 is for.

6 Experiment 2: Modern Sentence-Embedding Models

Setup and caveat. Three 2025 and 2026 embedding models that top retrieval leaderboards: Qwen3-Embedding-0.6B, Qwen3-Embedding-4B, and multilingual-E5-large-instruct. These produce one pooled vector per input, so at NC level we encode the *isolated phrase* (*grey matter* on its own vs *brain* on its own) rather than a contextual span, a difference to keep in mind when comparing to the encoders and LLMs.

Model	ISC	LOD	FLOOR	ICS
Qwen3-Embedding-4B	0.300	0.043	0.509	**0.478**
mE5-large-instruct	0.316	0.091	0.834	0.460
Qwen3-Embedding-0.6B	0.201	0.149	0.543	0.401

Table 7: Modern embedding models on English NCIMP (isolated-phrase level). The best, Qwen3-Embedding-4B (0.478), slightly exceeds the strongest 2024 encoder (E5-large, 0.438) but stays under 0.55.

Reading, with example. Two different model families converge on the same wall: Qwen3-Embedding-4B (0.478) and the decoder Qwen3-4B (0.479, next section) land at essentially the same score, both with near-zero lexical bias yet both below threshold. *Example of the residual failure:* at ISC 0.30 the gold synonym for an idiom is, on average, only 30% of the way above the random floor, a weak bridge rather than real understanding. Reducing lexical bias is not, by itself, enough; and a healthy low floor (Qwen3-Embedding's 0.51) is not enough either.

7 Experiment 3: The Decoder-Only LLM Cohort

Setup. Eleven models across two generations and both training stages: Qwen3 (2025) and Qwen3.5 (2026); Gemma 3 (2025) and Gemma 4 (2026); base and instruction-tuned; 0.8B to 12B. All at NC (contextual-span) level with the last-four-layers-plus-final-norm recipe of Section 4. Table 8 is generated directly from the run outputs, so the numbers and the text cannot drift apart.

Model	Gen.	Type	ISC	LOD	FLOOR	ICS
Qwen3.5-0.8B-Base	2026	base/mlx	+0.186	+0.101	0.561	0.425
Qwen3.5-2B-it	2026	instruct/mlx	+0.200	+0.060	0.511	0.444
Gemma3-1B-it	2025	instruct/mlx	+0.207	+0.080	0.667	0.455
Qwen3.5-9B-it	2026	instruct/mlx	+0.217	+0.010	0.540	0.478
Qwen3-4B-Base	2025	base/torch	+0.188	+0.050	0.825	0.479
Qwen3.5-4B-it	2026	instruct/mlx	+0.222	+0.011	0.523	0.479
Gemma4-E2B-Base	2026	base/torch	+0.213	+0.025	0.812	0.490
Gemma3-12B-pt	2025	base/mlx	+0.258	-0.006	0.556	0.498
Gemma3-4B-pt	2025	base/mlx	+0.254	+0.008	0.682	0.499
Gemma4-12B-it	2026	instruct/mlx	-0.006	+0.010	0.949	0.520
Gemma4-12B-base	2026	base/mlx	+0.260	-0.033	0.531	0.525

Table 8: Decoder-only LLMs on English NCIMP (mean over contexts), sorted by ICS. Threshold = 0.55. The instruction-tuned Gemma 4-12B is discussed below; its high ICS is an anisotropy artefact, not genuine capture (note its FLOOR of 0.95).

Three findings, each with an example.

1. **Nothing crosses** 0.55**.** The whole cohort sits at ICS 0.42 to 0.525, in the same band as the 2024 encoders of Experiment 1.
2. **The failure mode shifts.** Encoders fail by over-weighting literal pieces (LOD > 0); decoder LLMs push LOD to about zero. *Example:* the cohort leader, base Gemma 4-12B, reaches LOD $= -0.03$, the first model to put *brain* closer to *grey matter* than *silvery material*, the right preference at last. Yet its ISC is only 0.26 and its ICS 0.525, still short of the bar.
3. **Scale, generation, and alignment all approach but do not cross.** Within Qwen3.5, ICS saturates from about 4B onward at the 2025 Qwen3-4B value. In a like-for-like base comparison at 12B, the newer Gemma 4 (0.525) modestly beats Gemma 3 (0.498), so a newer generation helps a little, but the gain is small and below threshold.

The anisotropy artefact, concretely. Instruction-tuned Gemma 4-12B reports ICS 0.520, which looks competitive, but its FLOOR is 0.95: alignment tuning has collapsed its space so that everything is about 0.95 similar. Its genuine synonym recovery (ISC) is near 0, and the high composite is the metric instability of Section 3 (dividing by $1 - 0.95$), not real capture. The base version of the same model (ISC 0.26, FLOOR 0.53, ICS 0.525) is the real result. This is the second methodological finding: report FLOOR alongside ICS, always.

8 Experiment 4: The Ordinary-Perturbation Control

The missing control. Experiments 1 to 3 show models failing on idioms. But high similarity after a small edit could be a general property of sentence embeddings, not something about idioms. The original protocol never checks this, because its random baseline exists only for NC targets. We add the missing comparison: run the exact same synonym-versus-random test on non-idiomatic targets.

The controls, by example. Two new groups (16 items each) join the idiomatic and compositional compounds.

- **Ordinary single words**: *doctor* to *physician* (synonym) vs *doctor* to *carpet* (random).
- **Ordinary two-word phrases**: *red car* to *crimson vehicle* vs *red car* to *coffee spoon*.

There is no idiomaticity here, so any competent model should easily place the synonym above the random replacement.

The calibrated metric. For a group g, let Δ_g be its mean synonym-minus-random gap. The *ordinary-calibrated gap* expresses the idiomatic separation as a fraction of the ordinary one,

$$\mathrm{OCG}_g = \frac{\Delta_g}{\Delta_{\text{ordinary}}}, \qquad \Delta_{\text{ordinary}} = \tfrac{1}{2}(\Delta_{\text{ord. word}} + \Delta_{\text{ord. phrase}}).$$

OCG $= 1$ means idioms separate as cleanly as ordinary targets; OCG far below 1 means the synonym-random distinction specifically collapses for idioms.

Expected vs. actual, on one model. For mBERT at contextual-span level, the ordinary word *doctor* separates cleanly (*physician* $= 0.93$ vs *carpet* $= 0.77$, a gap of 0.16), while the idiomatic group gap is -0.01 (the gold synonym is, if anything, slightly farther than a random word). So mBERT's idiomatic OCG is essentially zero: the very same model that easily tells *physician* from *carpet* cannot tell an idiom's gold synonym from noise.

Sentence level is the wrong place to look. Before the span-level result, the control also explains a trap. At full-sentence level, random replacements stay highly similar for *every* group, because the shared sentence frame dominates the embedding (Figure 3). So a high sentence-level cosine is not idiomaticity evidence at all; the contextual span is where the distinction lives.

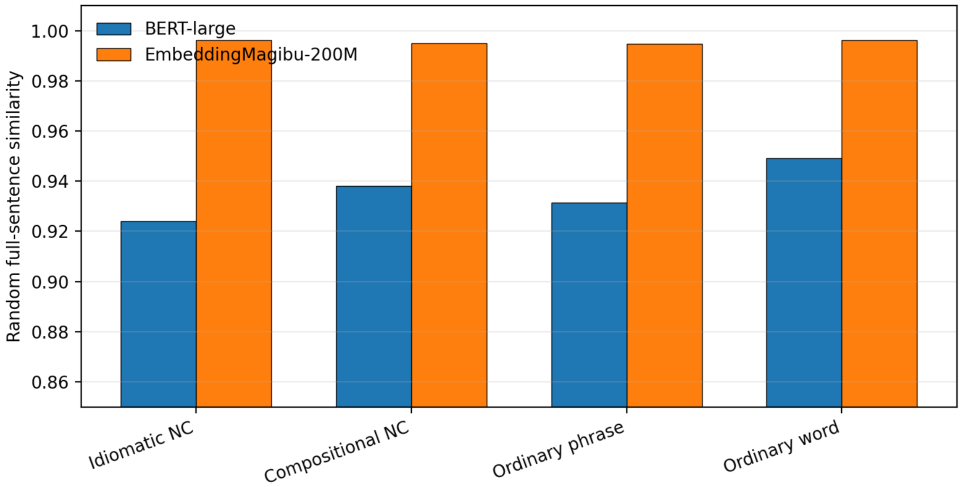

Figure 3: Random-replacement *full-sentence* similarity stays high (0.90 to 0.99) for all four groups, idiomatic and ordinary alike. The shared sentence frame, not idiomaticity, drives this; it is why the analysis must move to the contextual span.

Result across all twenty models. Figure 4 and Table 9 give the contextual-span OCG for the entire study cohort. The result is uniform: $\mathrm{OCG}_{\mathrm{idiom}}$ never reaches 1 (best: base Gemma 4-12B at 0.68), while the compositional calibrated gap sits at or above 1 for nearly every model. Compositional compounds behave like ordinary phrases; idioms do not. The paper-era encoders are worst, at about 0 or negative.

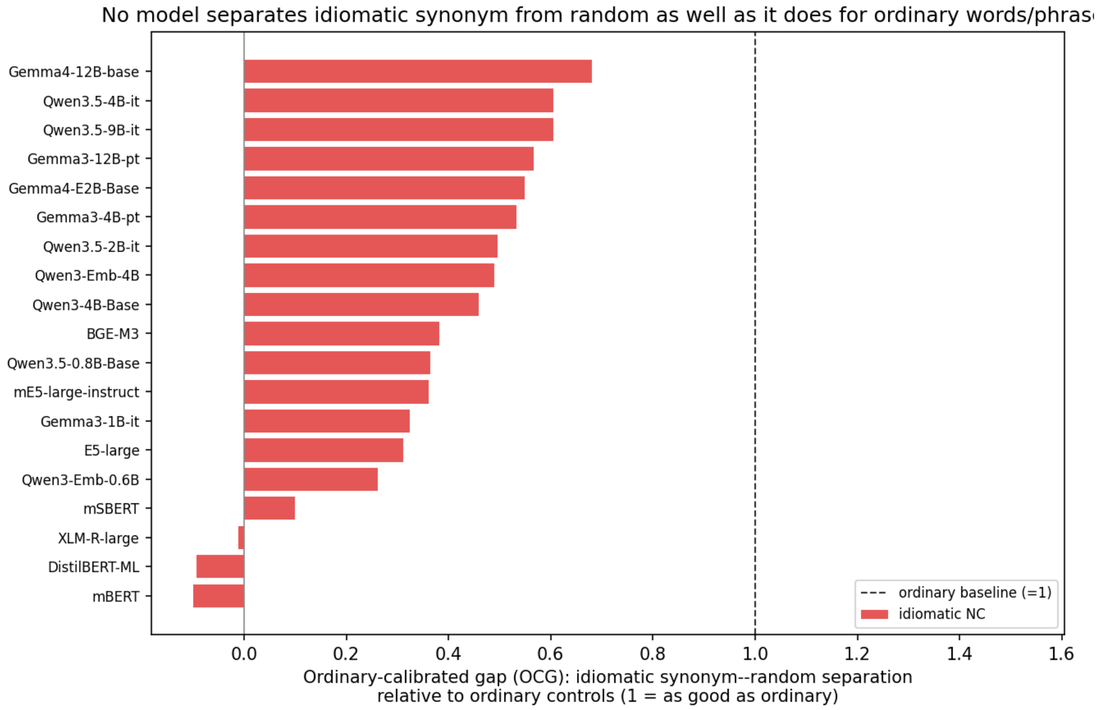

Figure 4: Ordinary-calibrated idiomatic gap (OCG) for all twenty models at contextual-span level. The dashed line at 1 is the ordinary-control baseline. Every model falls short: the newest base LLM (Gemma 4-12B) is closest at 0.68, the 2024 encoders sit near or below zero. Instruction-tuned Gemma 4-12B is omitted (its collapsed space makes the ratio unstable).

Model	idiom gap	ord. gap	OCG	Model	idiom gap	ord. gap	OCG
Gemma4-12B-base	0.186	0.273	**0.68**	Qwen3.5-0.8B-base	0.092	0.252	0.36
Qwen3.5-4B-it	0.147	0.242	0.61	mE5-large-instruct	0.042	0.115	0.36
Qwen3.5-9B-it	0.138	0.229	0.61	Gemma3-1B-it	0.074	0.227	0.33
Gemma3-12B-pt	0.155	0.273	0.57	E5-large	0.027	0.088	0.31
Gemma4-E2B-base	0.066	0.121	0.55	Qwen3-Emb-0.6B	0.070	0.267	0.26
Gemma3-4B-pt	0.109	0.205	0.53	mSBERT	0.053	0.533	0.10
Qwen3.5-2B-it	0.121	0.245	0.50	XLM-R-large	-0.000	0.037	-0.01
Qwen3-Emb-4B	0.139	0.284	0.49	DistilBERT-ML	-0.011	0.115	-0.09
Qwen3-4B-Base	0.054	0.117	0.46	mBERT	-0.012	0.124	-0.09
BGE-M3	0.097	0.254	0.38	Gemma4-12B-it*	0.001	0.003	0.20

Table 9: Ordinary-calibrated idiomatic gap for all twenty models (contextual span), sorted by OCG. The compositional gap (not shown) is at or above the ordinary baseline for nearly all models; *Gemma4-12B-it has a collapsed space, so its ratio is unstable.

For the two-model pilot that motivated this control (BERT-large and a local embedding model), the same picture holds at finer granularity (Figure 5): ordinary words and phrases reach near the baseline, idiomatic compounds reach only a fraction of it.

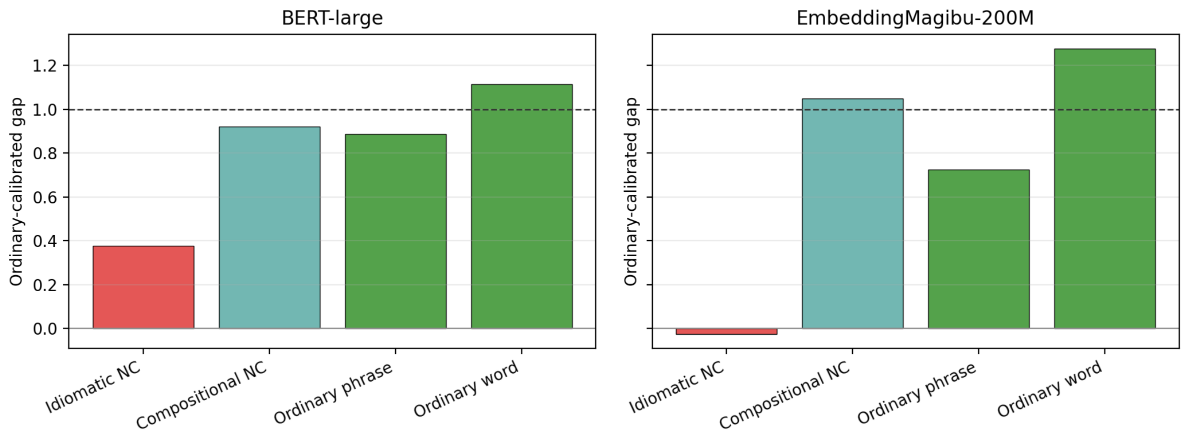

Figure 5: The original two-model pilot. BERT-large separates ordinary controls well but idiomatic compounds reach only about a third of the ordinary baseline; the local embedding model fails even more strongly on idioms. This pilot motivated running the calibration on all twenty models.

Reading. This is the cleanest single result in the study: the failure is specific to idiomaticity (compositional and ordinary targets separate fine), and even the best modern model recovers only two-thirds of the ordinary synonym-random separation for idioms.

9 Putting It Together: All Twenty Models

Table 10 collects the headline NC-level capture score for every model, across the three modelling families, in one place. The range is narrow (0.32 to 0.525) and the threshold line at 0.55 is never reached. Reading down the column, neither family, nor scale, nor year, nor training stage pulls clearly ahead.

Model	Family	Year/type	ICS
Gemma4-12B-base	decoder LLM	2026 base	0.525
Gemma4-12B-it*	decoder LLM	2026 instruct	0.520
Gemma3-4B-pt	decoder LLM	2025 base	0.499
Gemma3-12B-pt	decoder LLM	2025 base	0.498
Gemma4-E2B-base	decoder LLM	2026 base	0.490
Qwen3.5-4B-it	decoder LLM	2026 instruct	0.479
Qwen3-4B-Base	decoder LLM	2025 base	0.479
Qwen3.5-9B-it	decoder LLM	2026 instruct	0.478
Qwen3-Embedding-4B	embedding	2025	0.478
mE5-large-instruct	embedding	2024	0.460
Gemma3-1B-it	decoder LLM	2025 instruct	0.455
Qwen3.5-2B-it	decoder LLM	2026 instruct	0.444
E5-large	encoder/emb	2024	0.438
BGE-M3	encoder/emb	2024	0.428
Qwen3.5-0.8B-base	decoder LLM	2026 base	0.425
XLM-R-large	encoder	2024	0.424
Qwen3-Embedding-0.6B	embedding	2025	0.401
mBERT	encoder	2024	0.353
DistilBERT-ML	encoder	2024	0.349
mSBERT	encoder/emb	2024	0.320

Table 10: Every model in the study, English NC level, sorted by ICS. No model reaches the partial-capture threshold of 0.55. *Gemma4-12B-it's score is an anisotropy artefact (FLOOR 0.95); its base counterpart at the top of the table is the genuine leader.

10 Discussion

Reading the four experiments together gives a consistent, calibrated answer to "have newer models learned to read idioms?"

- **A soft ceiling, approached from every direction.** More parameters (0.8B to 12B), a newer generation (2024 to 2026), a different family (encoder, embedding, decoder), and alignment tuning all push toward, but none past, an ICS of about 0.50 to 0.53. The strongest model, base Gemma 4-12B, is the first to get the qualitative sign right (it prefers the gold synonym) yet still stops at 0.525.
- **The failure is specifically idiomatic.** The ordinary control rules out the deflating alternative that embeddings are simply insensitive to small edits: they separate *doctor* to *physician*/*carpet*, and even compositional compounds, cleanly, and fail only on idioms.
- **The bottleneck looks structural, not a matter of capacity.** If scale were the issue, the 12B models would pull away; they do not. If the representation merely needed de-biasing, the embedding models (low lexical bias) would cross the bar; they do not. What remains is a small surviving synonym-random margin for idioms that newer models widen only slightly. Likely routes forward are therefore not bigger models but idiom-aware objectives, adapters or curricula, and anisotropy-corrected representations.
- **Two methodological lessons.** For decoder LLMs the final layer's residual stream is anisotropic and must be normalised, or scores are noise (Section 4); and the composite ICS must be read with FLOOR, since extreme anisotropy inflates it spuriously (Section 7).

11 Limitations

The ordinary-control set is small (64 hand-built items) and some random replacements are grammatically awkward; a larger, balanced version would strengthen the calibration. The composite ICS is a heuristic for ranking, not ground truth; the per-indicator numbers (ISC, LOD, and the OCG of Experiment 4) carry the argument. The decoder-LLM cohort is English-only at NC level; extending it to Portuguese, where Experiment 1 already diverges from English, is the natural next step. Finally, MLX reports the final hidden state, so its recipe (last four layers plus final norm) is close to, but not identical with, the torch path; we keep base, full-precision weights to minimise other confounds.

12 Conclusion

We rebuilt the idiomaticity probe of He et al. (2025) as a model-agnostic framework and ran it on twenty models from 2024 encoders to 2026 LLMs, adding an MLX path for the newest architectures and a final-norm recipe that makes decoder-LLM representations measurable. The headline is a soft ceiling: no model captures idiomaticity, the best (base Gemma 4-12B) reaches only 0.525 while becoming the first to prefer the gold synonym over the literal one, and scale, generation, and alignment all approach but do not cross the bar. An ordinary-perturbation control on all twenty models shows the failure is specific to idioms: models separate *doctor* to *physician* from *carpet* cleanly but cannot do the same for an idiom's gold synonym. The original conclusion therefore stands, now sharpened and calibrated: representing what an idiom *means*, rather than what its words *say*, remains an open problem that newer and larger models have narrowed but not solved.

A Reproducibility: What Was Run

Every result in this paper is produced by the model-agnostic framework and saved under `runs/` and `experiments/perturbation_control/`. Table 11 maps each experiment to its data, models, and outputs. All twenty models were run offline from local caches; the article tables and figures are regenerated by a single asset-builder script, and the decoder-LLM table is written directly from the run outputs (so it cannot drift from the text).

Experiment	Models / data	Outputs
1. Encoders	6 models, EN+PT, NCIMP	`runs/en_all`, `runs/pt_all`
2. Embeddings	3 models, EN, NCIMP	`runs/emb_*`
3. Decoder LLMs	11 models, EN, NCIMP	`runs/llm_*`, `runs/small_*`
4. Ordinary control	20 models, 64-item control set	`experiments/perturbation_control/all_models`

Table 11: Experiment-to-output map. Detailed per-experiment write-ups, each with worked examples, are in `experiments_docs/`.

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